Pratikshya Regmi

Raleigh, NC 27606

Email: pregmi3@ncsu.edu ;

Website: https://pratikshya-regmi.github.io/

EDUCATION

- **Ph.D.** - In Progress

  Center For Geospatial Analytics, North Carolina State University, 2025.
  - Major: Geospatial Analytics
  - Dissertation *Innovative Methodology for Accurate Landscape Modelling using High Resolution Multisensor Datasets.*
    - Advisor: Dr. Helena Mitasova
- **M.S.**

  Geospatial Systems Engineering, Texas A & M University- Corpus Christi, 2023.
  - Thesis: *Application of UAS photogrammetry and geospatial AI techniques for palm tree detection and mapping*
    - Advisor: Dr. Michael J. Starek
- **BE**

  Geomatics Engineering, Kathmandu University, School of Engineering, Dhulikhel, Kavre 2018.
  - Thesis: *Analyzing the Effect of Image Overlap, Number of GCPs & its Distribution Pattern on Overall Accuracy of UAV Photogrammetry*
    - Advisor: Dr. Uma Shankar Pandey

POSITIONS AND AFFILIATIONS

- **Graduate Research and Teaching Assistant, North Carolina State University (2023-Present)**
  - GeoForAll lab Center for Geospatial Analytics.
    - Processing post-Helene imagery; developing LiDAR-UAS data-fusion models for accurate terrain modeling; working on satellite imagery; building a geospatial-LLM assistant; contributing to open-source geospatial software's (GRASS).
- **Research Intern, Oak Ridge National Laboratory (ORNL) (May 2026-Aug 2026)**
  - Ecosystem Processes Group.
    - Assessing Forest Resistance, Delayed Canopy Loss, and Early Recovery Following Hurricane Helene
- **Summer Research Assistant, North Carolina State University (June 2024-Aug 2024)**
  - Institute for transportation and Research education.
- **Graduate Research Assistant, Texas A &M University- Corpus Christi (TAMUCC) (2021-2023)**
  - MANTIS lab, Conrad Blucher Institute for Science and Technology
- **Geomatics Engineer, NEA Engineering Company Limited, Kathmandu Nepal (Jan 2021-July 2021)**

HONORS AND AWARDS

- Women in GIS Award- Women in GIS. 2025
- University Graduate Fellowship (Extraordinary incoming PhD student) - North Carolina State University. 2023
- C.J. Davidson Scholarship - Texas A&M University-Corpus Christi. 2023
- C.J. Davidson Scholarship - Texas A&M University-Corpus Christi. 2022
- Conrad Blucher Institute Scholarship - Conrad Blucher Institute. 2022
- Geospatial Surveying Engineering Scholarship - Texas A&M University-Corpus Christi. 2022
- ASPRS Student Conference Presentation Grant - American Society for Photogrammetry and Remote Sensing. 2022
- ·C.J. Davidson Scholarship- Texas A&M University–Corpus Christi. 2021
- Geospatial Surveying Engineering Scholarship Texas A&M University-Corpus Christi. 2021
- Best Paper Presented - Unmanned Aerial System in Geomatics Conference, IIT Roorkee, India · Jan 2019.

PUBLICATIONS AND PRESENTATIONS

Regmi, P., Mitasova, H. (2024). Visibility Analysis for Safe UAS Operation Using Lidar Datasets. Presented at the American Geophysical Union (AGU) Annual Meeting.

Regmi, P. (2023). *Application of UAS Photogrammetry and Geospatial AI Techniques for Palm Tree Detection and Mapping*. Master's Thesis, Texas A&M University-Corpus Christi.

Regmi, P., Starek, M. J., & Berryhill, J. (2023). Overview of the Uncrewed Aircraft System (UAS) Campus Survey Project at Texas A&M University-Corpus Christi. FIG Working Week 2023.

Regmi, P., Starek, M. J., & Berryhill, J. (2022). A Historical Perspective of UAS Monitoring for Campus Facilities and Geoinformatics at Texas A&M University-Corpus Christi. ASPRS 2022 Annual Conference.

Regmi, P., & Wagle, N. (2021). Encroachment of Forest Cover in Chitwan and Parsha Districts of Nepal: A Spatiotemporal Quantification Over 25 Years. Journal of Land Management and Geomatics Education, 3, 7–12.

Awasthi, B., Karki, S., **Regmi, P**., Dhami, D. S., Thapa, S., & Panday, U. S. (2020). Analyzing the Effect of Distribution Pattern and Number of GCPs on Overall Accuracy of UAV Photogrammetric Results. In Proceedings of UASG 2019: Unmanned Aerial System in Geomatics, 1, 339–354.

Awasthi, B., Karki, S., **Regmi, P**., Dhami, D. S., Thapa, S., & Panday, U. S. (2019). Assessing the Overall Accuracy of UAV Photogrammetry: A Methodology for Different Combination of Imagery Parameters and Terrain Types. Journal of Land Management and Geomatics Education, 1, 22–32.

WORKSHOPS

Hosted two-hour clinic at CSDMS annual meeting 2025(Intro to GRASS GIS and Tangible Landscape-2025, UC Boulder, Colorado) https://cnr.ncsu.edu/geospatial/news/2025/06/23/center-ph-d-students-engage-global-audience-with-grass-gis-and-tangible-landscape-workshop/

Hosted 2-hour GRASS GIS clinic at CSDMS annual meeting (Coastal evolution analysis and inundation modeling with GRASS GIS using Jupyter Notebook- Montclair State University, New Jersey) https://cnr.ncsu.edu/geospatial/news/2024/06/03/students-teach-grass-gis/

HACKATHON

SAFE-MCP- 2025- By Aastha.ai https://www.linkedin.com/posts/safe-mcp

NC PSI Hackathon- 2025 – **Advanced Track Winner https://cnr.ncsu.edu/geospatial/news/**

NC PSI HACKATHON- 2024- Beginners Track (Participant)

PRESENTATIONS

- Assessment of Landscape Disturbance and Recovery After Hurricane Helene Using Multi-Temporal Elevation and Imagery Data (IALE NA 2026, Athens, Georgia)

- Geomorphic Impacts and Terrain Recovery After Hurricane Helene: Multi- Sensor Geospatial Analysis in Western North Carolina (AGU 2025, New Orleans, LA)

- Accessing the Potential of Large Language Models (LLMs) for Geospatial Analysis (IALE NA 2025, Raleigh, North Carolina)

- Visibility Analysis for Safe UAS Operation (AGU 2024, Washington D.C.)

- Simulating Wildfire Ash Transport Following a Precipitation Event by Coupling Dorado and Overland Flow (CSDMS Fall Webinar Series- Virtual)

- Overview of the Uncrewed Aircraft System (UAS) Campus Survey Project at Texas A&M University-Corpus Christi (Presented at FIG Working Week 2023, Orlando, Florida)

- Application of Uncrewed Aircraft Systems (UAS) Surveying and Geospatial AI to Detect and Monitor Palm Trees (TAMUCC Symposium, 2023)

- A Historical Perspective of UAS Monitoring for Campus Facilities and Geo-informatics at Texas A&M University-Corpus Christi (Student Presentation Grant on ASPRS Annual Conference 2022, Denver Colorado)

- Analyzing the Effect of Distribution Pattern and Number of GCPs on Overall Accuracy of UAV Photogrammetric Results (Best paper presented on UASG-2019, IIT, Roorkee, UP, India)

TECHNICAL SKILLS

- Programming (Python, R)
- Geospatial Software's (GRASS, Arcgis pro, ArcGIS online, QGIS)
- Image Processing (Pix4D, Agisoft Metashape, Cloud Compare)
- Cloud Computing (Google Earth Engine)
- Other skills (MATLAB, latex,)

PROFESSIONAL SERVICE

- Nepali Student Association (NSA) NCSU (President, 2026-Present)
- AGU H3S Sub Committee (Member, 2025-2026)
- Nepali Student Association (NSA) NCSU (Vice President, 2025-2026)
- Geospatial Graduate Student Organization (GGSO) (External Affairs Officer, 2024-2025)
- American Geophysical Union (AGU) (Member, 2024-Present)
- American Society for Photogrammetry and Remote Sensing (ASPRS) (Member, 2023-2024)

VOLUNTEERING

STEM RESOURCE FAIR FOR K-12 STUDENTS WITH DISABILITIES AT NORTH CAROLINA STATE UNIVERSITY (2024 AND 2025) Featured an interactive activity where students used blocks to learn about satellite imagery and its wide applications.

CEETLE EDUCATION, SCHOOL OF APPLIED SOFTWARE ENGINEERING AND LEADERSHIP (TEACHING ASSISTANT) | DEC 2020- FEB 2021 Teaching the fundamentals of data science and machine learning using streaming data from Array of Things (AOT) sensors to high school students of Nepal who are eager to learn but could not afford.

LICENCES AND CERTIFICATIONS

- Thriving in Interdisciplinary Teams workshop, Shelton Leadership Institute (March 2026)
- Leadership Learning Institute, Graduate School, NCSU (March 2026)
- AUVSI Trusted Operator, AUVSI. May 2025
- FAA Part 107 Certified sUAS Remote Pilot, FAA. Jul 2024
- Spatial Data Science: The New Frontier in Analytics, Esri. Nov 2022
- ArcGIS Notebooks Basics, Esri. Jun 2022
- Data Science Workflows Using ArcGIS Notebooks, Esri. Jun 2022
- Introduction to ArcGIS API for Python, Esri. Jun 2022

- Introduction to Regression Analysis Using ArcGIS Pro, Esri. Jun 2022
- Mapping Clusters: Hot Spot & Outlier Analysis, Esri. Nov 2021
- Crime Analysis: Tactical and Strategic, Esri. Oct 2021
- Optimal Facility Location Using ArcGIS Pro, Esri. Oct 2021
- Introduction to the Crime Analysis Solution, Esri. Oct 2021
- Space-Time Analysis: Time-Series Clustering, Esri. Sep 2021
- Capstone: Retrieving, Processing & Visualizing Data with Python, Univ. of Michigan. Feb 2021
- Machine Learning, Stanford University (Andrew Ng). Oct 2020
- Neural Networks and Deep Learning, DeepLearning.AI (Andrew Ng). Oct 2020
- Python Data Structures, University of Michigan. Oct 2020
- Python for Everybody: Getting Started, University of Michigan. Sep 2020